

EARTH AND SPACE



Year 10 Earth and Space Unit 1

The universe contains features including galaxies, stars and solar systems, and the Big Bang theory can be used to explain the origin of the universe ([ACSSU188](#))

Week	Topic	Outcomes	Resources	Assessment (plus weighting)
1	Introduction to the universe	• Students to review prior learning • Size of the universe and scale • Measuring distance in space • Parallax and how to use it measure distances in space • Light years as a measure of distance • Calculating light speed	Stile app Education Perfect Universe intro video Scale app Microwave and chocolate melting	
2	Big Bang Theory	• Introduction to the Big Bang Theory • Evidence supporting the Big Bang • Rapid expansion Theory • Scientists and theories supporting the Big Bang	Google classroom Big Bang Rubric Stile app	Big Bang Theory report (7.5%)
3	Life cycle of Stars	• stellar life cycles (small and large stars) • Planet formation from the remains of exploded stars • Classification of galaxies • Evidence for the Big Bang • Hubbles Law, redshift, Doppler effect	Stile app https://www.twig-world.com/film/what-are-stars-913/ Netflix - the origins of the universe.	
4	Observing the night sky	• Use of technology to observe universe • Radio telescopes, satellites etc. and how they are used to record the observable universe • Magnitude of brightness to determine the size of stars • EM spectrum and satellites		
5	Neutron stars and black holes Review and consolidate	• death of stars and what they become • Black holes, gravity and gravitational waves	Bookcreator revision resource	Topic test (5%) Glossary Semester 1 exam (20%)

Year 10 Earth and Space Unit 2

Global systems, including the carbon cycle, rely on interactions involving the biosphere, lithosphere, hydrosphere and atmosphere (ACSSU189)

Week	Topic	Outcomes	Resources	Assessment (plus weighting)
1	Global Systems	Looking at the different closed system cycles on earth - carbon, nitrogen and water cycles	- stile app - Twig video	
2	Global Systems	- examples of how the carbon, nitrogen and water cycles work in closed systems, how biotic and abiotic factors affect each cycle. (phosphorus cycle)	Keynote	
3	Climate history and change	- What is the greenhouse effect? - The positive and negatives of a greenhouse effect - Climate change and sea levels/biodiversity. How is the climate changing and how does human activity affect it - Temperature change over time.	www	research report on Human influenced climate change (7.5%)
4	Human effects on earth systems	- climate change - the effects of climate change (Australian droughts, floods, hurricanes etc) - Permafrost and ice loss in the Tundra and polar regions of earth - Habitat loss as a cause of climate change		
5	Biodiversity	- loss of biodiversity and its effects on ecosystems - Habitat loss		- topic test (5%) - Semester 2 exam (20%)

